

LAPORAN BAB 2

Praktikum Devops

Konsep Container dan Instalasi Docker



Disusun Oleh:

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PROGRAM REKOGNISI PEMBELAJARAN LAMPAU (RPL)

JURUSAN TEKNIK INFORMATIKA

POLITEKNIK ELEKTRONIKA NEGERI SURABAYA

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1 Tujuan dan Ruang Lingkup

Praktikum Bab 2 bertujuan untuk:

1. Menjelaskan perbedaan virtual machine dan container dari sisi isolasi, ukuran, startup time, dan overhead.
2. Mengidentifikasi komponen Docker: client, daemon, registry, image, container, network, dan volume.
3. Menginstal Docker Engine pada Ubuntu serta memahami struktur instalasi pada Windows berbasis WSL2.
4. Menjalankan container pertama, melakukan inspeksi image, melihat log, dan membangun image custom sederhana.

Ruang lingkup bab ini mencakup instalasi Docker Engine (client, daemon, containerd, buildx, dan compose plugin), uji docker run hello-world, eksperimen container Nginx dan Ubuntu interaktif, serta pembangunan image custom web statis berbasis nginx:1.26-alpine.

2 Pelaksanaan Praktikum

2.1 Instalasi Docker Engine pada Ubuntu

- 1) Sudo apt update

: Memperbarui daftar paket aplikasi dari repositori Ubuntu.

```
ubuntu123@DESKTOP-7A8ST00:~$ sudo apt update
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [601 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [142 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.9 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [268 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [88.3 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [407 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [79.5 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Get:13 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [471 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [109 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [171 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [53.8 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.5 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [397 kB]
Get:21 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [77.8 kB]
Fetched 3654 kB in 12s (295 kB/s)
148 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu123@DESKTOP-7A8ST00:~$
```

- 2) sudo apt install -y ca-certificates curl gnupg lsb-release

: Menginstal paket pendukung yang dibutuhkan untuk mengunduh, mengamankan, dan memverifikasi sumber instalasi Docker.

```
ubuntu123@DESKTOP-7A8ST00:~$ sudo apt install -y ca-certificates curl gnupg lsb-release
curl is already the newest version (8.18.0-1ubuntu2.4).
curl set to manually installed.
lsb-release is already the newest version (12.1-2build1).
lsb-release set to manually installed.
Upgrading:
  ca-certificates  dirnmgr  gnupg  gnupg-l10n  gnupg-utils  gpg  gpg-agent  gpg-wks-client  gpgconf  gpgsm

Summary:
  Upgrading: 10, Installing: 0, Removing: 0, Not Upgrading: 138
  Download size: 2261 kB
  Freed space: 91.1 kB

Get:1 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 ca-certificates all 20260601~26.04.1 [139 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 gnupg-utils amd64 2.4.8-4ubuntu3.1 [127 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 gpg-wks-client amd64 2.4.8-4ubuntu3.1 [71.9 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 dirnmgr amd64 2.4.8-4ubuntu3.1 [327 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 gpg-agent amd64 2.4.8-4ubuntu3.1 [236 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 gpg amd64 2.4.8-4ubuntu3.1 [637 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 gpgconf amd64 2.4.8-4ubuntu3.1 [106 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 gnupg all 2.4.8-4ubuntu3.1 [311 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 gpgsm amd64 2.4.8-4ubuntu3.1 [239 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 gnupg-l10n all 2.4.8-4ubuntu3.1 [68.8 kB]
Fetched 2261 kB in 2s (984 kB/s)
Preconfiguring packages ...
```

- 3) sudo install -m 0755 -d /etc/apt/keyrings

: Membuat direktori khusus untuk menyimpan kunci keamanan (GPG key).

```
ubuntu123@DESKTOP-7A8ST00:~$ sudo install -m 0755 -d /etc/apt/keyrings
ubuntu123@DESKTOP-7A8ST00:~$ |
```

- 4) curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \

sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg

: Mengunduh kunci GPG resmi milik Docker dan menyimpannya agar sistem meyakini bahwa paket Docker yang diunduh aman dan asli.

```
ubuntu123@DESKTOP-7A8ST00:~$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \
  sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
ubuntu123@DESKTOP-7A8ST00:~$ |
```

- 5) sudo chmod a+r /etc/apt/keyrings/docker.gpg

: Memberikan izin akses baca pada file kunci GPG tersebut.

```
ubuntu123@DESKTOP-7A8ST00:~$ sudo chmod a+r /etc/apt/keyrings/docker.gpg
ubuntu123@DESKTOP-7A8ST00:~$
```

- 6) echo "deb [arch=\$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \

https://download.docker.com/linux/ubuntu \$(lsb_release -cs) stable" | \

sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

: Menambahkan repositori resmi Docker ke dalam daftar sumber aplikasi Ubuntu Anda. (Catatan: Pada teks Anda ada sedikit kesalahan baris/spasi antara docker.gpgecho, yang seharusnya dipisah menjadi dua perintah terpisah).

```
ubuntu123@DESKTOP-7A8ST00:~$ echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \
  https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
ubuntu123@DESKTOP-7A8ST00:~$ |
```

- 7) Sudo apt update

: Memperbarui kembali daftar paket agar sistem mengenali repositori Docker yang baru saja ditambahkan.

```
ubuntu123@DESKTOP-7A8ST00:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 https://download.docker.com/linux/ubuntu resolute InRelease [32.5 kB]
Get:5 https://download.docker.com/linux/ubuntu resolute/stable amd64 Packages [24.9 kB]
Hit:6 http://security.ubuntu.com/ubuntu resolute-security InRelease
Fetched 57.4 kB in 11s (5383 B/s)
138 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu123@DESKTOP-7A8ST00:~$
```

- 8) `sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin`
: Menginstal komponen utama Docker, meliputi Engine (ce), Command Line Interface (cli), container runtime (containerd), serta plugin buildx dan docker-compose.

```
ubuntu123@DESKTOP-7A8ST00:~$ sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Installing:
  containerd.io  docker-buildx-plugin  docker-ce  docker-ce-cli  docker-compose-plugin

Installing dependencies:
  docker-ce-rootless-extras  libip4tc2  libnetfilter-contrack3  libnftables1  nftables
  iptables  libip6tc2  libnftnl1  libnftnl11  pigz

Suggested packages:
  cgroupfs-mount | cgroup-lite  docker-model-plugin  firewallld

Summary:
  Upgrading: 0, Installing: 15, Removing: 0, Not Upgrading: 138
  Download size: 101 MB
  Space needed: 386 MB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libip4tc2 amd64 1.8.11-2ubuntu3 [24.2 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libip6tc2 amd64 1.8.11-2ubuntu3 [24.4 kB]
update-alternatives: using /usr/sbin/ip6tables-nft to provide /usr/sbin/ip6tables (ip6tables) in auto mode
update-alternatives: using /usr/sbin/arptables-nft to provide /usr/sbin/arptables (arptables) in auto mode
update-alternatives: using /usr/sbin/ebtables-nft to provide /usr/sbin/ebtables (ebtables) in auto mode
Setting up docker-ce (5:29.8.0-1~ubuntu.26.04~resolute) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' → '/usr/lib/systemd/system/docker.service'.
Created symlink '/etc/systemd/system/sockets.target.wants/docker.socket' → '/usr/lib/systemd/system/docker.socket'.
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
ubuntu123@DESKTOP-7A8ST00:~$
```

- 9) `sudo usermod -aG docker $USER`
: Memasukkan user Anda saat ini ke dalam grup docker.

```
ubuntu123@DESKTOP-7A8ST00:~$ sudo usermod -aG docker $USER
ubuntu123@DESKTOP-7A8ST00:~$
```

- 10) `newgrp docker`
: Mengaktifkan perubahan grup secara langsung pada sesi terminal yang sedang berjalan tanpa perlu *logout* terlebih dahulu.

```
ubuntu123@DESKTOP-7A8ST00:~$ newgrp docker
ubuntu123@DESKTOP-7A8ST00:~$
```

- 11) `docker version`
: Menampilkan informasi versi Docker untuk memastikan aplikasi sudah terpasang.

```
ubuntu123@DESKTOP-7A8ST00:~$ docker version
Client: Docker Engine - Community
 Version:           29.8.0
 API version:       1.56
 Go version:        go1.26.8
 Git commit:        88096ef
 Built:             Thu Sep  3 21:50:20 2026
 OS/Arch:           linux/amd64
 Context:           default

Server: Docker Engine - Community
 Engine:
  Version:           29.8.0
  API version:       1.56 (minimum version 1.40)
  Go version:        go1.26.8
  Git commit:        3ce5872
  Built:             Thu Sep  3 21:50:20 2026
  OS/Arch:           linux/amd64
  Experimental:      false
 containerd:
  Version:           v2.3.4
  GitCommit:         db8809540e1a7a9da5d518876894933ff55692ab
 runc:
  Version:           1.5.1
  GitCommit:         v1.5.1-0-g8f2685a4
 docker-init:
  Version:           0.19.0
  GitCommit:         de40ad0
ubuntu123@DESKTOP-7A8ST00:~$ |
```

12) docker run hello-world

: Mengunduh dan menjalankan container uji coba sederhana untuk memastikan Docker dapat berfungsi dengan baik.

```
ubuntu123@DESKTOP-7A8ST00:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbdb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

2.2 Container Nginx dan Ubuntu Interaktif

- 1) docker pull nginx:1.26

: Mengunduh *image* Nginx versi 1.26 dari Docker Hub ke dalam komputer lokal.

```
ubuntu123@DESKTOP-7A8ST00:~$ docker pull nginx:1.26
1.26: Pulling from library/nginx
4fd410795c0f: Pull complete
8a628cdd7ccc: Pull complete
7a0654aeb922: Pull complete
6923759e66ab: Pull complete
9ebfb40fb06b: Pull complete
5e98d206134b: Pull complete
d44088bb6ae8: Pull complete
f17adcc09044: Download complete
5c2c3686e536: Download complete
Digest: sha256:41b194461e4bae16f9b25d68b0976ed4735b89ca625c89aad88e1c1c3b7e8860
Status: Downloaded newer image for nginx:1.26
docker.io/library/nginx:1.26
ubuntu123@DESKTOP-7A8ST00:~$ |
```

- 2) docker run -d --name web-public -p 8080:80 nginx:1.26

: Menjalankan container dari image nginx:1.26 di latar belakang (-d), memberi nama container web-public, dan menyambungkan port 8080 di komputer Anda ke port 80 milik container.

```
ubuntu123@DESKTOP-7A8ST00:~$ docker run -d --name web-public -p 8080:80 nginx:1.26
f7c2c33c8213f20ed7c5f9bd2daa1d65b2832a4057b26c6ca76c4bac6e840729
ubuntu123@DESKTOP-7A8ST00:~$ |
```

- 3) docker ps

: Menampilkan daftar container yang sedang aktif berjalan.

```
ubuntu123@DESKTOP-7A8ST00:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                    NAMES
f7c2c33c8213   nginx:1.26 /docker-entrypoint..." 27 seconds ago Up 26 seconds    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web-public
ubuntu123@DESKTOP-7A8ST00:~$
```

- 4) `docker logs --tail 20 web-public`

: Melihat 20 baris log/catatan aktivitas terakhir dari container web-public.

```
ubuntu123@DESKTOP-7A8ST00:~$ docker logs --tail 20 web-public
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.
sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/09/07 13:20:57 [notice] 1#1: using the "epoll" event method
2026/09/07 13:20:57 [notice] 1#1: nginx/1.26.3
2026/09/07 13:20:57 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)
2026/09/07 13:20:57 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/09/07 13:20:57 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
2026/09/07 13:20:57 [notice] 1#1: start worker processes
2026/09/07 13:20:57 [notice] 1#1: start worker process 29
2026/09/07 13:20:57 [notice] 1#1: start worker process 30
2026/09/07 13:20:57 [notice] 1#1: start worker process 31
2026/09/07 13:20:57 [notice] 1#1: start worker process 32
2026/09/07 13:20:57 [notice] 1#1: start worker process 33
2026/09/07 13:20:57 [notice] 1#1: start worker process 34
2026/09/07 13:20:57 [notice] 1#1: start worker process 35
2026/09/07 13:20:57 [notice] 1#1: start worker process 36
2026/09/07 13:20:57 [notice] 1#1: start worker process 37
2026/09/07 13:20:57 [notice] 1#1: start worker process 38
2026/09/07 13:20:57 [notice] 1#1: start worker process 39
2026/09/07 13:20:57 [notice] 1#1: start worker process 40
ubuntu123@DESKTOP-7A8ST00:~$
```

- 5) `curl http://localhost:8080`

: Mengirimkan permintaan ke server Nginx di port 8080 untuk memastikan halaman web utama Nginx berhasil diakses dari komputer Anda.

```
ubuntu123@DESKTOP-7A8ST00:~$ curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
ubuntu123@DESKTOP-7A8ST00:~$
```

- 6) `docker run -it --name ubuntu-test ubuntu:22.04 /bin/bash`
: Menjalankan container Ubuntu 22.04 secara interaktif (-it) dengan nama ubuntu-test dan langsung masuk ke dalam terminal shell-nya (/bin/bash).

```
ubuntu123@DESKTOP-7A8ST00:~$ docker run -it --name ubuntu-test ubuntu:22.04 /bin/bash
Unable to find image 'ubuntu:22.04' locally
22.04: Pulling from library/ubuntu
d544298cabd5: Pull complete
5370ab2c86d9: Download complete
Digest: sha256:2edbbc5dc405e9612ba3584ce95480277e3eb374407b5505fe26f17df77c7dbc
Status: Downloaded newer image for ubuntu:22.04
root@0ee58f10b10d:/# |
```

- 7) `cat /etc/os-release`
: Menampilkan informasi versi sistem operasi di dalam terminal Ubuntu container yang sedang aktif tersebut.

```
root@0ee58f10b10d:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
root@0ee58f10b10d:/# |
```

- 8) Exit
: Keluar dari terminal container Ubuntu dan menghentikan jalannya container tersebut.
- 9) `docker rm -f web-public ubuntu-test`
: Menghapus secara paksa (-f) kedua container (web-public dan ubuntu-test) dari sistem.

```
ubuntu123@DESKTOP-7A8ST00:~$ docker rm -f web-public ubuntu-test
web-public
ubuntu-test
ubuntu123@DESKTOP-7A8ST00:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
ubuntu123@DESKTOP-7A8ST00:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED      STATUS      PORTS      NAMES
66d04ee974df   hello-world  "/hello"   8 hours ago  Exited (0)  8 hours ago  goofy_proskuriakova
ubuntu123@DESKTOP-7A8ST00:~$ |
```

2.3 Pembangunan Image Custom Web Statis

- 1) `mkdir -p ~/docker-lab/custom-web && cd ~/docker-lab/custom-web`
: Membuat direktori kerja baru dan langsung masuk ke dalamnya.

```
ubuntu123@DESKTOP-7A8ST00:~$ ls
devsecops-lab
ubuntu123@DESKTOP-7A8ST00:~$ mkdir -p ~/docker-lab/custom-web && cd ~/docker-lab/custom-web
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/custom-web$ |
```

- 2) `cat > index.html << 'EOF'` □ Titik awal input text

: Membuat file index.html yang berisi konten HTML sederhana.

```
<h1>Docker Lab PENS</h1>
```

```
<p>Container berhasil berjalan.</p>
```

EOF □ Titik Akhir

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/custom-web$ cat > index.html << 'EOF'
> <h1>Docker Lab PENS</h1>
> <p>Container berhasil berjalan.</p>
> EOF
```

- 3) cat > Dockerfile << 'EOF' □ Titik awal input □ Titik awal input

FROM nginx:1.26-alpine

: Menggunakan base image Nginx versi Alpine yang ringan.

LABEL maintainer=admin@pens.ac.id

: Menambahkan informasi pembuat/pengelola image.

COPY index.html /usr/share/nginx/html/index.html

: Menyalin file HTML buatan kita ke dalam direktori web Nginx.

EXPOSE 80

: Menandai port 80 sebagai port layanan.

CMD ["nginx", "-g", "daemon off;"]

: Perintah utama untuk menjalankan Nginx di *foreground*.

EOF □ Titik akhir input text

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/custom-web$ cat > Dockerfile << 'EOF'
> FROM nginx:1.26-alpine
> LABEL maintainer="admin@pens.ac.id"
> COPY index.html /usr/share/nginx/html/index.html
> EXPOSE 80
> CMD ["nginx", "-g", "daemon off;"]
> EOF
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/custom-web$
```

- 4) docker build -t pens-web:1.0 .

: Membangun image baru bernama pens-web:1.0 dari Dockerfile di direktori saat ini.

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/custom-web$ docker build -t pens-web:1.0 .
[+] Building 22.0s (8/8) FINISHED          docker:default
=> [internal] load build definition from Dockerfile      0.0s
=> => transferring dockerfile: 192B                      0.0s
=> [internal] load metadata for docker.io/library/nginx  3.7s
=> [auth] library/nginx:pull token for registry-1.docker 0.0s
=> [internal] load .dockerignore                        0.0s
=> => transferring context: 2B                            0.0s
=> [internal] load build context                        0.0s
```

- 5) docker run -d --name pens-app -p 9090:80 pens-web:1.0

: Menjalankan container di latar belakang (-d) dengan nama pens-app, mengarahkan port 9090 komputer ke port 80 container.

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/custom-web$ docker run -d --name pens-app -p 9090:80 pens-web:1.0
8ae61443a6387b0a8da4051aa2ab0269de64f4eed9dd59a6fc0adee57b2547f2
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/custom-web$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
8ae61443a638   pens-web:1.0   "/docker-entrypoint..." 15 seconds ago Up 14 seconds 0.0.0.0:9090->80/tcp, [::]:9090->80/tcp
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/custom-web$
```

6) curl <http://localhost:9090>

: Memeriksa apakah halaman web kustom berhasil diakses melalui port 9090.

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/custom-web$ curl http://localhost:9090
<h1>Docker Lab PENS</h1>
<p>Container berhasil berjalan.</p>
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/custom-web$
```

3 Analisis Wajib

- Jelaskan satu masalah yang muncul dan cara Anda mendiagnosisnya.
 - Masalah: Pada Langkah 2.2 Point 4, saat memeriksa docker logs web-public, Nginx spawn hingga 11 worker process (worker process 30 hingga 40).
 - Diagnosis: Hal ini terjadi karena konfigurasi default Nginx (worker_processes auto;) membaca seluruh virtual CPU core yang dialokasikan oleh Hyper-V/WSL2 dari host Windows. Untuk container web server sederhana di lingkungan pengujian, alokasi worker process yang terlalu banyak ini membuang-buang memori (resource waste).
- Jelaskan risiko keamanan atau operasional yang relevan pada bab ini.
 - Keamanan:
 - Root Privilege: Container berjalan menggunakan akun default root bawaan image nginx:1.26-alpine yang berisiko jika terjadi container escape.
 - Akses Grup Docker: Penambahan pengguna ke grup docker (usermod -aG docker) memberikan akses setara root pada host melalui soket Docker.
 - Lalu Lintas Unencrypted: Layanan HTTP terbuka pada port unencrypted (80/8080/9090) tanpa TLS/SSL.
 - Operasional:
 - Resource Spikes: Tidak adanya batasan CPU/RAM pada perintah docker run berisiko memicu crash pada host jika terjadi memory leak.
 - Penghentian Paksa: Penggunaan docker rm -f berpotensi memutus proses/sesi yang sedang berjalan secara tidak aman (non-graceful shutdown).
- Berikan rekomendasi perbaikan bila lab ini akan dibawa ke production-like environment
 - User Non-Root: Konfigurasi Dockerfile agar layanan berjalan di bawah ID pengguna non-root (misal USER 1001).
 - Resource Limit: Tambahkan parameter pembatas alokasi memori dan CPU saat deployment (contoh: --memory="256m" --cpus="0.5").

- Enkripsi & TLS: Gunakan Reverse Proxy (Nginx/Traefik) dengan sertifikat SSL/TLS untuk enkripsi HTTPS.
- Scan Image (DevSecOps): Integrasikan pemindaian kerentanan image (seperti Trivy atau Grype) pada CI/CD pipeline.

4 Kesimpulan

Praktikum Bab 2 menyelesaikan instalasi Docker Engine pada Ubuntu, menjalankan container pertama, melakukan inspeksi image, melihat log, dan membangun image custom. Hasil memperlihatkan bahwa container adalah mekanisme isolasi proses yang berbagi kernel host, sehingga ringan dan cepat, tetapi boundary keamanannya bergantung pada konfigurasi namespace, cgroup, capabilities, dan kebijakan akses daemon. Penggunaan tag spesifik, proses non-root, penghindaran secret pada layer, serta pencatatan ingest dibuktikan sebagai praktik dasar yang menjadi fondasi untuk eksperimen network, storage, dan Compose pada bab berikutnya. Bukti eksekusi, kendala yang ditemui, dan refleksi keamanan/operasional terdokumentasi pada bagian verifikasi dan analisis.